Depth Anything V2: Depth Estimation and Numerical Analysis

1. Introduction

In this exercise, the **Depth Anything V2 Small** model was used to estimate the depth of an RGB image from the **Ultralytics Depth8 dataset**[1]. Depth8 is a compact dataset containing eight indoor RGB-D images sampled from the SUN RGB-D dataset. It is designed primarily for testing and debugging monocular depth estimation pipelines rather than benchmarking models. The dataset contains paired RGB images and ground-truth depth maps with real indoor depth measurements.

The selected image was image_01.png.

2. Depth Estimation

The input RGB image was loaded using PIL and converted to RGB format. The pretrained model depth-anything/Depth-Anything-V2-Small-hf was loaded using Hugging Face Transformers.

The model receives an RGB image as input. After preprocessing, the image is converted into a tensor suitable for the model. The model then produces a two-dimensional depth prediction corresponding to the spatial locations of the input image.

The predicted depth map was resized to the original image dimensions and displayed alongside the RGB image. It was also saved as an image file and as a NumPy array.

The resulting depth map had the following dimensions: **Depth shape:** (427, 561)

Thus, each pixel in the RGB image has a corresponding predicted depth value.

3. Numerical Analysis

The following statistics were calculated for the predicted depth map:

Property	Result
Shape	(427, 561)
Data type	float32
Minimum value	0.1424
Maximum value	6.5203
Mean	3.1419
Standard deviation	1.3730

[1] Depth8 Dataset – Ultralytics Documentation

The minimum predicted value was **0.1424**, located at:

(row, column) = (234, 542)

The maximum predicted value was **6.5203**, located at:

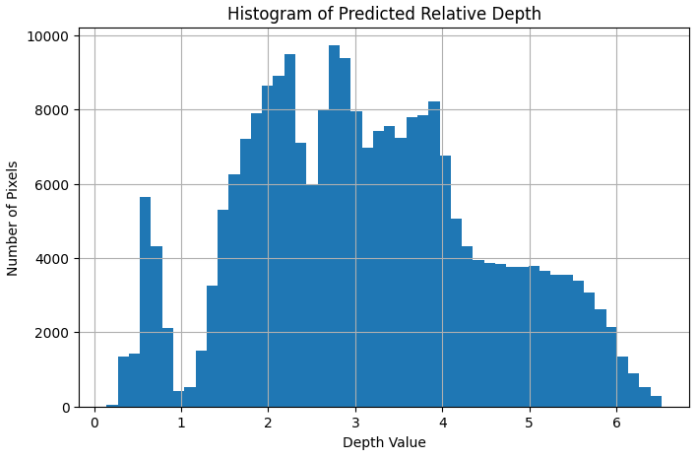
(row, column) = (267, 0)

The center pixel was located at:

(row, column) = (213, 280)

and its predicted depth value was **4.1182**.

The depth histogram was also plotted to show the distribution of predicted depth values across all pixels. The histogram describes how frequently different relative depth levels occur in the image. A wide distribution indicates greater variation in estimated depth across the scene.



4. Relative Depth and Metric Depth

An important point is that the output of a monocular depth estimation model should not automatically be interpreted as a physical distance in meters. The prediction represents **relative depth** rather than directly calibrated metric depth.

In addition, the raw prediction can be interpreted in terms of **inverse depth/disparity**, where larger values generally correspond to closer regions and smaller values to farther regions. Therefore, the numerical value itself does not represent a distance measurement.

For example, a predicted value of 0.25 does **not** mean that the corresponding object is exactly 25 cm from the camera.

Metric depth, in contrast, represents physical distance and is normally expressed in units such as meters. Converting a relative/inverse-depth prediction into metric depth requires appropriate scale information or calibration.

Therefore:

depth = 0.25 ≠ 0.25 meters

and it cannot be interpreted as 25 cm without additional metric calibration.

5. Conclusion

In this exercise, Depth Anything V2 was successfully applied to an RGB image from the Depth8 dataset. A corresponding depth map was generated, visualized, saved, and numerically analyzed.

The predicted map had a resolution of 427×561 with float32 values ranging from 0.1424 to 6.5203. The results demonstrate that monocular depth estimation can provide useful information about the relative spatial structure of a scene. However, the predicted values should not be interpreted directly as physical distances because the model output represents relative/inverse depth rather than calibrated metric depth.

The Depth8 dataset is particularly suitable for this type of exercise because it provides paired RGB and depth data, although its eight-image size makes it unsuitable for meaningful benchmarking of depth-estimation models.